



Fig. 6: Data mining—a useful tool for various types of complex financial analysis (Source: [www.marketoracle.co.uk](http://www.marketoracle.co.uk))



Fig. 7: UPS, the world's largest courier company receives 39.5 million tracking requests per day (Source: [www.mb.com.ph](http://www.mb.com.ph))

National Crime Record Bureau (NCRB) maintains a large crime database and uses crime data mining techniques such as clustering. In earlier times, extracting and exchanging information among police agencies was a very time-consuming process and often data was not available in the time of need.

The government of India made use of IT and implemented a government-to-government (G2G) model called crime criminal information system (CCIS). This system is designed to create computerised storage, analysis and retrieval of criminal records. For example, if a person has been robbed of his or her mobile phone, the police can use data mining tools to search the IMEI number on the basis of the mobile phone number and get it blocked.

**Healthcare.** Healthcare transactions generate a vast amount of data,

and this too is multi-faceted and large to be processed and analysed by conventional means. Data mining can advance decision making by discovering patterns and trends in large amounts of complex data.

In healthcare, major application areas include evaluation of treatment effectiveness, management of healthcare, customer relationship management (CRM), and detection of fraud and abuse.

By comparing and contrasting causes, symptoms and courses of treatments, data mining can provide an analysis of the courses of action that can prove to be successful. For example, results of patient groups treated with various drug regimens for the same ailment can be compared to establish which treatment is most effective and offers higher value for money.

**Financial applications.** Data mining is a very useful tool in this sector, and forecasting stock markets, currency exchange rates, bank bankruptcies, understanding and managing financial risks, trading futures, credit ratings, loan management, bank customer profiling and money-laundering analyses are some core tasks.

Cross-selling refers to selling a range of related products to a single customer profitably, based on certain assumptions. These assumptions are based on understanding products that have synergy with customer profiles and their requirements.

Banks often attempt to cross-sell credit cards, pensions and life insurance policies. Standard Chartered Grindlays Bank faced severe competition from other banks due to poaching, similarity in products and services and other factors. Its business intelligence unit using data analysis and a test-and-learn culture was able to find out the likelihood of customers to take on a new product. For example, it knew which of its card members were more likely to take an auto loan, resulting in more focused marketing campaigns and reduced costs with improved customer satisfaction. As a result, the marketing department was empowered with information to increase cross-holding, target most valuable customers and also help in the next best product strategy for a customer.

It is expected that there will be a high growth of hybrid methods, that is, those that merge diverse models and deliver better performance vis-à-vis individuals in the area of data mining in finance.

In this integrative approach, individual models work like trained artificial experts. Hence, their combinations can be oriented in the same manner as consultation by human experts. Also, these artificial experts can be successfully pooled with human experts. In times to come, these artificial experts will be configured as autonomous intelligent software agents.

**Sales and marketing applications.** Data mining is used to arrive at a customer's value, also known as lifetime value (LTV), which is a useful concept in measuring customer retention. Data mining can also be used to predict a customer's likelihood to switch to competition. Probability scores for each customer can